Pinecones

01 What is a pinecone made of?



Scales stack in a spiral around a pinecone.

Look closer. Follow a spiral around the cone.

A little explanation. A pinecone is a woody seed case with many overlapping parts called scales. Seeds grow tucked beside the scales. The scales wind around the cone in spirals.

Try this. Count the scales you can see on one side, then draw one spiral.

02 Why are there gaps in a dry cone?



Dry scales open, leaving room for seeds to leave.

Look closer. Look between the spread-out scales.

A little explanation. When many common pinecones dry, their scales bend outward. The open spaces let ripe seeds leave. A light wing helps many pine seeds catch the breeze.

Try this. Set a dry cone on a dish and sketch its outline. Leave room beside your drawing for a second shape.

03 What changes when a cone gets wet?



Wet scales bend inward and close the gaps.

Look closer. Compare the close scales with your open-cone drawing.

A little explanation. Wet or humid conditions add water to the scales. Parts of each scale take in water differently, so the woody tissue bends inward. The cone closes. A fallen cone can still do this without muscles.

Try this. Use two fallen cones or cones from a craft bin. Put one in a dry dish and briefly wet the other in a shallow dish of water. With Big beside you, check their scales over time and draw both shapes. Collect fallen brown cones with Big nearby; leave green cones on branches, keep feet on the ground, and wash hands afterward.

04 Why wait for dry air?



Dry weather can help winged seeds drift away.

Look closer. Find the thin wing beside a seed.

A little explanation. Winged seeds can drift on a dry breeze. When wet, they can cling to a scale instead. Closing in damp weather helps keep seeds inside until conditions are better for leaving. Different kinds of pinecones do not all open in exactly the same way.

Try this. Let the damp cone dry on an empty dish. Check it later. What do you think the scales will do?